

Plus-configuration quadcopter: explicit rigid-body dynamics

Explicit equations of the model implemented as `QuadcopterBox` / `QuadcopterBox2` in `dynamics/dynamics.py`, and equivalently as the CASADI/ACADOS model `Model.f_exp1`. The two implementations were verified to agree to machine precision (max absolute discrepancy $5.7 \cdot 10^{-14}$ over $2 \cdot 10^4$ random states and inputs); the equations below were themselves re-derived independently and checked against the code (max discrepancy $2.8 \cdot 10^{-14}$).

1 Notation and state

Shorthand: $c_\phi = \cos \phi$, $s_\phi = \sin \phi$, $c_\theta = \cos \theta$, $s_\theta = \sin \theta$, $t_\theta = \tan \theta$, $c_\psi = \cos \psi$, $s_\psi = \sin \psi$.

$$\mathbf{x} = \left(\underbrace{p_x, p_y, p_z}_{\text{position}}, \underbrace{\phi, \theta, \psi}_{\text{roll, pitch, yaw}}, \underbrace{v_x, v_y, v_z}_{\text{linear velocity}}, \underbrace{\omega_x, \omega_y, \omega_z}_{\text{body rates}}, \underbrace{b}_{\text{box half-width}} \right) \in \mathbb{R}^{13} \quad (1)$$

$$\mathbf{u} = (u_1, u_2, u_3, u_4) \in \mathbb{R}^4, \quad u_i \in [0, \bar{u}] \quad (2)$$

Position and linear velocity are expressed in the world frame; body rates $\boldsymbol{\omega} = (\omega_x, \omega_y, \omega_z)$ in the body frame. Each u_i is the thrust of rotor i in newtons.

2 Rotation matrix

ZYX Euler convention, body $\rightarrow$ world:

$$\mathbf{R}(\phi, \theta, \psi) = \mathbf{R}_z(\psi) \mathbf{R}_y(\theta) \mathbf{R}_x(\phi) = \begin{bmatrix} c_\psi c_\theta & c_\psi s_\theta s_\phi - s_\psi c_\phi & c_\psi s_\theta c_\phi + s_\psi s_\phi \\ s_\psi c_\theta & s_\psi s_\theta s_\phi + c_\psi c_\phi & s_\psi s_\theta c_\phi - c_\psi s_\phi \\ -s_\theta & c_\theta s_\phi & c_\theta c_\phi \end{bmatrix} \quad (3)$$

with

$$\mathbf{R}_x = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c_\phi & -s_\phi \\ 0 & s_\phi & c_\phi \end{bmatrix}, \quad \mathbf{R}_y = \begin{bmatrix} c_\theta & 0 & s_\theta \\ 0 & 1 & 0 \\ -s_\theta & 0 & c_\theta \end{bmatrix}, \quad \mathbf{R}_z = \begin{bmatrix} c_\psi & -s_\psi & 0 \\ s_\psi & c_\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (4)$$

Only the third column of $\mathbf{R}$ enters the dynamics, since the total thrust acts along the body z axis:

$$\mathbf{R} \mathbf{e}_3 = \begin{bmatrix} c_\phi s_\theta c_\psi + s_\phi s_\psi \\ c_\phi s_\theta s_\psi - s_\phi c_\psi \\ c_\phi c_\theta \end{bmatrix} \quad (5)$$

3 Wrench produced by the rotors

Plus (+) configuration: rotor 1 at $+x_B$, rotor 2 at $+y_B$, rotor 3 at $-x_B$, rotor 4 at $-y_B$, all thrusts along $+z_B$.

$$\mathbf{f}_B = \begin{bmatrix} 0 \\ 0 \\ \sum_{i=1}^4 u_i \end{bmatrix}, \quad \boldsymbol{\tau} = \begin{bmatrix} \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} = \begin{bmatrix} \ell (u_2 - u_4) \\ \ell (u_3 - u_1) \\ \kappa (u_1 - u_2 + u_3 - u_4) \end{bmatrix}, \quad \kappa := \frac{c_t}{c_f} \quad (6)$$

In matrix form, $\mathbf{f}_B = \mathbf{F}\mathbf{u}$ and $\boldsymbol{\tau} = \mathbf{M}\mathbf{u}$ with

$$\mathbf{F} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}, \quad \mathbf{M} = \begin{bmatrix} 0 & \ell & 0 & -\ell \\ -\ell & 0 & \ell & 0 \\ \kappa & -\kappa & \kappa & -\kappa \end{bmatrix} \quad (7)$$

Roll is actuated by the pair (2, 4), pitch by the pair (3, 1); the alternating signs in τ_z are the rotor reaction torques, so hovering with $u_1 = u_2 = u_3 = u_4$ yields $\boldsymbol{\tau} = \mathbf{0}$ identically.

4 Equations of motion

4.1 Compact form

$$\dot{\mathbf{p}} = \mathbf{v} \quad (8)$$

$$\dot{\boldsymbol{\eta}} = \mathbf{T}^{-1}(\phi, \theta) \boldsymbol{\omega} \quad (9)$$

$$\dot{\mathbf{v}} = \frac{1}{m} \left(\sum_i u_i \right) \mathbf{R} \mathbf{e}_3 - g \mathbf{e}_3 \quad (10)$$

$$\mathbf{J} \dot{\boldsymbol{\omega}} = \boldsymbol{\tau} - \boldsymbol{\omega} \times (\mathbf{J} \boldsymbol{\omega}) \quad (11)$$

$$\dot{b} = 0 \quad (12)$$

with $\boldsymbol{\eta} = (\phi, \theta, \psi)$ and

$$\mathbf{T}^{-1}(\phi, \theta) = \begin{bmatrix} 1 & s_\phi t_\theta & c_\phi t_\theta \\ 0 & c_\phi & -s_\phi \\ 0 & \frac{s_\phi}{c_\theta} & \frac{c_\phi}{c_\theta} \end{bmatrix} \quad (13)$$

The inertia matrix is diagonal, $\mathbf{J} = \text{diag}(J_{xx}, J_{yy}, J_{zz})$, which is what makes the gyroscopic term expand into the scalar equations below.

4.2 Explicit component form

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}) = \begin{cases} \dot{p}_x = v_x \\ \dot{p}_y = v_y \\ \dot{p}_z = v_z \\ \dot{\phi} = \omega_x + s_\phi t_\theta \omega_y + c_\phi t_\theta \omega_z \\ \dot{\theta} = c_\phi \omega_y - s_\phi \omega_z \\ \dot{\psi} = \frac{s_\phi}{c_\theta} \omega_y + \frac{c_\phi}{c_\theta} \omega_z \\ \dot{v}_x = \frac{u_1 + u_2 + u_3 + u_4}{m} (c_\phi s_\theta c_\psi + s_\phi s_\psi) \\ \dot{v}_y = \frac{u_1 + u_2 + u_3 + u_4}{m} (c_\phi s_\theta s_\psi - s_\phi c_\psi) \\ \dot{v}_z = \frac{u_1 + u_2 + u_3 + u_4}{m} c_\phi c_\theta - g \\ \dot{\omega}_x = \frac{1}{J_{xx}} \left[\ell (u_2 - u_4) - \omega_y \omega_z (J_{zz} - J_{yy}) \right] \\ \dot{\omega}_y = \frac{1}{J_{yy}} \left[\ell (u_3 - u_1) - \omega_x \omega_z (J_{xx} - J_{zz}) \right] \\ \dot{\omega}_z = \frac{1}{J_{zz}} \left[\kappa (u_1 - u_2 + u_3 - u_4) - \omega_x \omega_y (J_{yy} - J_{xx}) \right] \\ \dot{b} = 0 \end{cases} \quad (14)$$

The last component is the box half-width: a static parameter carried inside the state so that a single value function generalises over box sizes.

5 Parameters

Symbol	Meaning	Value	Unit
m	mass	1.5	kg
ℓ	arm length (rotor to centre)	0.1	m
J_{xx}	inertia about x_B	0.0155	kg m ²
J_{yy}	inertia about y_B	0.0147	kg m ²
J_{zz}	inertia about z_B	0.0251	kg m ²
c_f	thrust coefficient	$1.5 \cdot 10^{-3}$	N/(rad/s) ²
c_t	drag-torque coefficient	$4.59 \cdot 10^{-5}$	N m/(rad/s) ²
κ	c_t/c_f	0.0306	m
g	gravity	9.81	m/s ²
$\bar{u}$	max thrust per rotor	17.496	N

Derived quantities: hovering thrust per rotor $mg/4 = 3.679$ N; thrust-to-weight ratio $4\bar{u}/(mg) = 69.98/14.72 \approx 4.76$.

6 Relation to the acados input parametrisation

The ACADOS model takes *squared rotor rates* ϖ_i^2 as inputs, whereas the equations above take thrusts in newtons. The two are related by the thrust coefficient:

$$u_i = c_f \varpi_i^2 \quad (15)$$

Consistently, its allocation matrices carry the coefficients explicitly,

$$\mathbf{F}^{\text{acados}} = c_f \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}, \quad \mathbf{M}^{\text{acados}} = \begin{bmatrix} 0 & \ell c_f & 0 & -\ell c_f \\ -\ell c_f & 0 & \ell c_f & 0 \\ c_t & -c_t & c_t & -c_t \end{bmatrix} \quad (16)$$

which reproduces $\mathbf{M}$ of Section 3 after the substitution, since $\ell c_f \varpi^2 = \ell u$ and $c_t \varpi^2 = (c_t/c_f)u = \kappa u$. The input bound matches as well: $\bar{u} = c_f \cdot 108^2 = 1.5 \cdot 10^{-3} \cdot 11664 = 17.496$ N.

7 Domain of validity

Equations (13) and (14) contain $\tan \theta$ and $1/\cos \theta$, so the Euler-angle kinematics are singular at $|\theta| = \pi/2$ (gimbal lock). In `float32` the singular factor reaches $\approx 1.3 \cdot 10^7$ there rather than overflowing, so a numerical integrator crossing that pitch produces very large but *finite* garbage rather than NaN: at $|\theta| = \pi/2 - 10^{-4}$ a single $\Delta t = 0.01$ s Euler step displaces ψ by about 500 rad. Note also that $\dot{\phi}$ and $\dot{\psi}$ share the singular factor, $\dot{\phi}|_{\text{sing}} = s_\theta \cdot \dot{\psi}|_{\text{sing}}$, so roll and yaw diverge together.

Two further modelling remarks:

- The expansion of $\boldsymbol{\omega} \times (\mathbf{J}\boldsymbol{\omega})$ in (14) assumes $\mathbf{J}$ diagonal. With off-diagonal inertia terms the compact form $\mathbf{J}\dot{\boldsymbol{\omega}} = \boldsymbol{\tau} - \boldsymbol{\omega} \times (\mathbf{J}\boldsymbol{\omega})$ still holds, but the scalar equations do not.
- No aerodynamic drag, rotor dynamics, or actuator lag is modelled: thrusts are treated as instantaneous inputs bounded in $[0, \bar{u}]$.